Health and Fitness Trends Analysis Using Python

1. Title Page

Item	Details
Project	VitalityApp Case Study
Title	**Health and Fitness Trends Analysis Using Python**
Student name	*[Insert student name]*
Course	*[Insert course name]*
Date	*[Insert submission date]*

2. Introduction

Health and fitness data analysis helps organizations understand how people move, monitor progress, and identify groups that may need additional support. For VitalityApp, daily steps are a practical indicator of physical activity, while height and weight provide basic body-measurement context. This case study uses Python to explore a small VitalityApp user sample and summarize activity patterns.

The analysis is descriptive: it reports the values observed in the available dataset and does not establish medical outcomes or causal relationships.

3. Problem Statement

VitalityApp needs a simple way to examine users' daily steps, height, weight, and activity level. Without analysis, it is difficult to see whether users are generally active, how many exceed a meaningful step threshold, or whether body measurements show an observable pattern in the sample.

The project therefore analyzes these measures to support fitness tracking, identify potentially less-active users, and provide evidence for engagement initiatives.

4. Objectives

The objectives of this analysis are to:

1. Identify descriptive fitness and activity trends in the available user sample.
2. Calculate the average daily step count.
3. Count users with more than 10,000 daily steps.
4. Compare active and inactive users by count and percentage.
5. Visualize the distribution of daily steps and the relationship between height and weight.

5. Tools and Technologies

Tool	Role in the project
Python	Main programming language for analysis and visualization.
Pandas	Loading, cleaning, summarizing, and filtering tabular data.
Matplotlib	Creating the histogram, scatter plot, and bar-chart visualizations.
Google Colab	Recommended cloud notebook environment for running the Python workflow.

6. Dataset Description

The available VitalityApp analysis dataset contains **20 users**. Each row represents one user and includes a user identifier, age, body measurements, daily steps, and an activity segment.

Column	Description	Observed use in this study
`Daily_Steps`	Number of steps recorded for a user in a day.	Used to calculate activity statistics and identify users above 10,000 steps.
`Height_cm`	User height in centimetres.	Used in the height–weight comparison.
`Weight_kg`	User weight in kilograms.	Used in the height–weight comparison.
`Active_User`	Boolean activity segment (`True` or `False`).	Used to compare active and inactive users.

In this dataset, `Active_User = True` represents the existing active segment definition, described as at least 8,000 daily steps. The observed daily-step values range from **3,162** to **14,567** steps.

7. Methodology

The analysis followed these steps:

1. **Data loading:** Read the dataset into a Pandas DataFrame.
2. **Exploration:** Reviewed columns, sample size, numeric ranges, and activity-segment values.
3. **Statistical analysis:** Calculated the mean daily steps, median, minimum, maximum, and count above 10,000 steps.
4. **Feature segmentation:** Used `Active_User` to group users into active and inactive segments.
5. **Visualization:** Used a histogram for daily-step distribution, a scatter plot for height versus weight, and a bar chart for activity-segment counts.

Example Python code

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the VitalityApp dataset
df = pd.read_parquet("fitness_measurements_active_users.parquet")

# Core daily-step statistics
average_steps = df["Daily_Steps"].mean()
users_over_10000 = (df["Daily_Steps"] > 10000).sum()

# Segment summary
segment_summary = (
    df["Active_User"]
    .value_counts()
    .rename(index={True: "Active", False: "Inactive"})
)

print(average_steps)
print(users_over_10000)
print(segment_summary)
```

> In Google Colab, the file path should be replaced with the location of the uploaded dataset or mounted Drive file.

8. Data Analysis and Results

8.1 Daily-step statistics

Measure	Result

Users analyzed	20
Average daily steps	10,080.8
Median daily steps	10,229
Minimum daily steps	3,162
Maximum daily steps	14,567
Users with more than 10,000 daily steps	10
Percentage with more than 10,000 daily steps	50.0%

The sample average is slightly above 10,000 daily steps. Exactly half of the users recorded more than 10,000 steps, indicating a mixed activity profile rather than a uniformly high or low one.

8.2 Active and inactive users

Activity segment	User count	Percentage	Mean daily steps
Active	14	70.0%	11,885.0
Inactive	6	30.0%	5,871.0
Total	**20**	**100.0%**	—

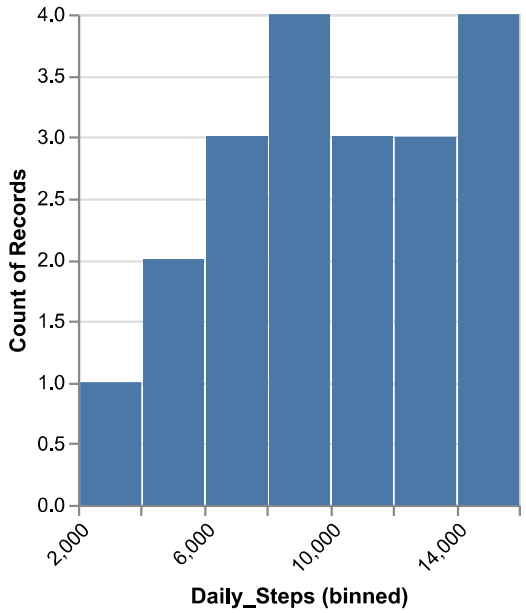
Most sampled users belong to the active segment. The active group also has a higher observed mean daily-step count than the inactive group, which is consistent with the segment definition.

8.3 Height and weight relationship

The height–weight scatter plot shows substantial variation in weight at similar heights. The Pearson correlation in this 20-user sample is **−0.28**, a weak negative association. This is a descriptive result only; the sample is too small and limited to support broad conclusions about body measurements.

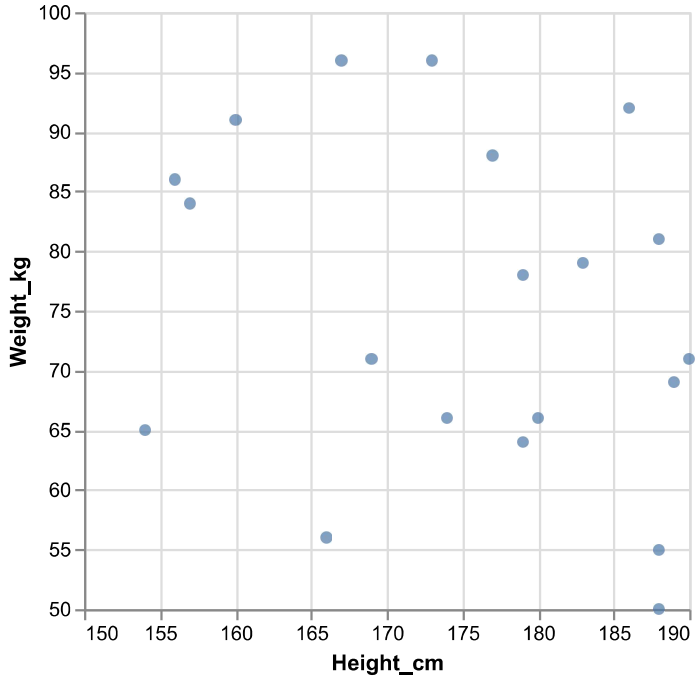
9. Visualizations

9.1 Distribution of daily steps



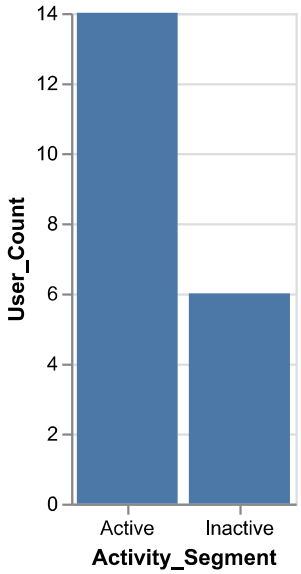
Interpretation: The histogram displays the spread of daily step counts across the 20 users. Values span from 3,162 to 14,567 steps, with the sample mean at 10,080.8 steps.

9.2 Height and weight comparison



Interpretation: The scatter plot compares `Height_cm` with `Weight_kg`. The points do not show a strong linear pattern in this sample; the calculated correlation is weakly negative (−0.28).

9.3 Activity-segment comparison



Interpretation: The bar chart shows that 14 users (70.0%) are active and 6 users (30.0%) are inactive. The inactive segment is smaller but may be relevant for targeted reminders, goals, or coaching features.

Example visualization code

```
# Histogram: daily-step distribution
plt.hist(df["Daily_Steps"], bins=6, edgecolor="black")
plt.title("Distribution of Daily Steps")
plt.xlabel("Daily Steps")
plt.ylabel("Number of Users")
plt.show()

# Scatter plot: height and weight
plt.scatter(df["Height_cm"], df["Weight_kg"])
plt.title("Height and Weight of VitalityApp Users")
plt.xlabel("Height (cm)")
plt.ylabel("Weight (kg)")
plt.show()
```

10. Key Findings

1. The dataset contains **20 users** with daily steps ranging from **3,162** to **14,567**.
2. Average daily steps are **10,080.8**, while the median is **10,229**.
3. **10 users (50.0%)** recorded more than 10,000 daily steps.
4. **14 users (70.0%)** are active and **6 users (30.0%)** are inactive according to the provided segment.
5. Active users average **11,885.0** daily steps, compared with **5,871.0** for inactive users.
6. Height and weight have a weak negative correlation (**r = −0.28**) in this small sample; no strong linear relationship is evident.

11. Conclusion

This VitalityApp case study demonstrates how Python can turn basic fitness records into useful descriptive insights. The results show that most sampled users are in the active segment and that half exceed 10,000 daily steps. The inactive group, representing 30% of the sample, could be a practical focus for engagement efforts such as reminders, achievable step goals, and progress feedback.

The analysis can help VitalityApp monitor activity patterns and communicate simple, evidence-based feedback to users. However, the dataset is a small one-time sample, so the findings should not be generalized to all users or used as medical advice.